1、review
2、inter segment
3、missed approach （alititude、waypoint、turn as soon as posible）

RNAV ILS



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Objectives

ADD,minus, mount , except

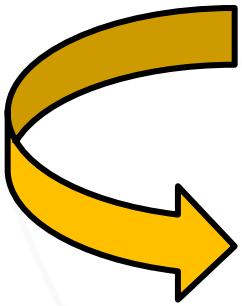
APP obs
Zgp'= （x+900） *tan3°
MPP obs H_{ma}
SOC= （OCH-HL)/TAN3-900

- List PBN applications usable for transition to ILS
- Know how to connect RNAV intermediate segment to ILS precision segment
- Know how to connect ILS precision segment to an RNAV missed approach with a Turning point (TP)
- Know how to compute OCH of missed appoach obstacle
  - Before E'
  - Between E' and earliest TP
  - Within the turn area of a missed approach with turn at TP
- Know how to connect ILS precision segment to an RNAV missed approach with a turn at an altitude and then DF to a waypoint
- Know how to compute OCH of missed appoach obstacles
  - For TH ≤ 300m
  - For TH ≥ 300m
- Identify the significant features for publication and coding

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RNAV ILS Concept

- Today :
  - Only ILS achieves DH ≤ 200 Ft
  - SBAS coverage is not worldwide and few airports are GBAS equipped
  - Major Airports are ILS equipped
  - Most of commercial aircraft are ILS equipped and RNAV1 capable
  - RNAV trajectories in terminal area or missed approach improve airspace concept (strategic separation, gain of flying distance, Merge Point , and environmental benefits)



The associated airspace concept to RNAV ILS is to:

- Take benefit of ILS guidance (cat II et III)
- Take benefit of RNAV flexibility
- Use the flexibility of RNAV track to mitigate radar failure
- Take benefit of commercial aircraft equipment on major airports

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WHICH PBN APPLICATION FOR TRANSITION TO ILS

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PBN Application up to IF for ILS procedure

Ref PatII section 1 § 1.21

- RNAV or RNP route with only systems capable of navigation accuracy of 1 NM or lower.
- PBN Applications:
  - ARNP (TSE ≤1),
  - RNAV1,
  - RNP1,
  - RNPAPCH (for initial and final missed approach)

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PBN application key parameters regarding procedure design criteria

Navigation Specification		Navaid Infrastructure	Leg Type	Flyby	RNP (NM)	TSE (NM)
RNP1	SID - STAR- Initial	GNSS	IF CF TF DF VA VM VI CA FA FM	Oui	1	≤ 1
RNAV1	SID - STAR- Initial	DMEDME GNSS		Oui	-	≤ 1
ARNP	SID - STAR- Initial	GNSS	+ RF + HM	Oui	1 à 0.3	≤ 1à 0.3
RNP APCH	Approach initial + intermediate + Final missed approach	GNSS	IF TF DF	Oui	1 0.3	≤ 1 ≤ 0.3

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TRANSITION RNAV ILS

How to connect RNAV intermediate segment to ILS precision segment

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Transition RNAV to ILS

At IF the system switches from RNAV mode to ILS mode

- Laterally : LOC capture
- When procedure is executed using managed mode (system), the anticipation is managed by system (even if turn angle at IF is wide)
- When using Flight Director , the risk of overshooting the LOC axis increases. Two issues have been identified :
  - No Anticipation made due to early LOC mode activation (No RNAV turn anticipation)
  - LOC display sensitivity high => on the display , LOC needle is active very late
- The Intermediate distance shall be long enough to intercept LOC then glide
  - With short distance => Glide interception by above

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INITIAL SEGMENT

- Navigation RNAV
  - Racetrack possible
    - Inbound = LOC guidance
    - Protection as holding
    - In France, we use secondary area = 1.25 NM
  - No reversal
- Ends at IF
- IF is aligned onto the LOC Axis
  - When distance IF/LOC ≥ 25 NM ⇒ Control flight
  - IF is defined as Fly By Waypoint (anticipation)
  - Max turn angle : 90°
  - Optimum : 30° (auto-pilot / LOC coupling issue)

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INTERMEDIATE SEGMENT LENGHT

Table II-1-1-1. Minimum distance between localizer and glide path interceptions

Intercept angle with localizer (degrees)	*Cat A/B/H*	*Cat C/D/E*
0 — 15	2.8 km (1.5 NM)	2.8 km (1.5 NM)
16 —30	3.7 km (2.0 NM)	3.7 km (2.0 NM)
31 — 60	3.7 km (2.0 NM)	4.6 km (2.5 NM)
61 — 90 or within a racetrack or reversal procedure	3.7 km (2.0 NM)	5.6 km (3.0 NM)

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DIFERENCE between RNAV transition to ILS and conventionnal navigation to ILS

NEW

- Missed approach OAS limits
- Connection intermediate / OAS
- RNAV guidance in missed approach (obstacles are assessed differently)

Unchanged

- **Standard conditions**
- **Height Loss**
- **Obstacle classification (approach or missed)**
- **Computation of obstacle equivalent height**

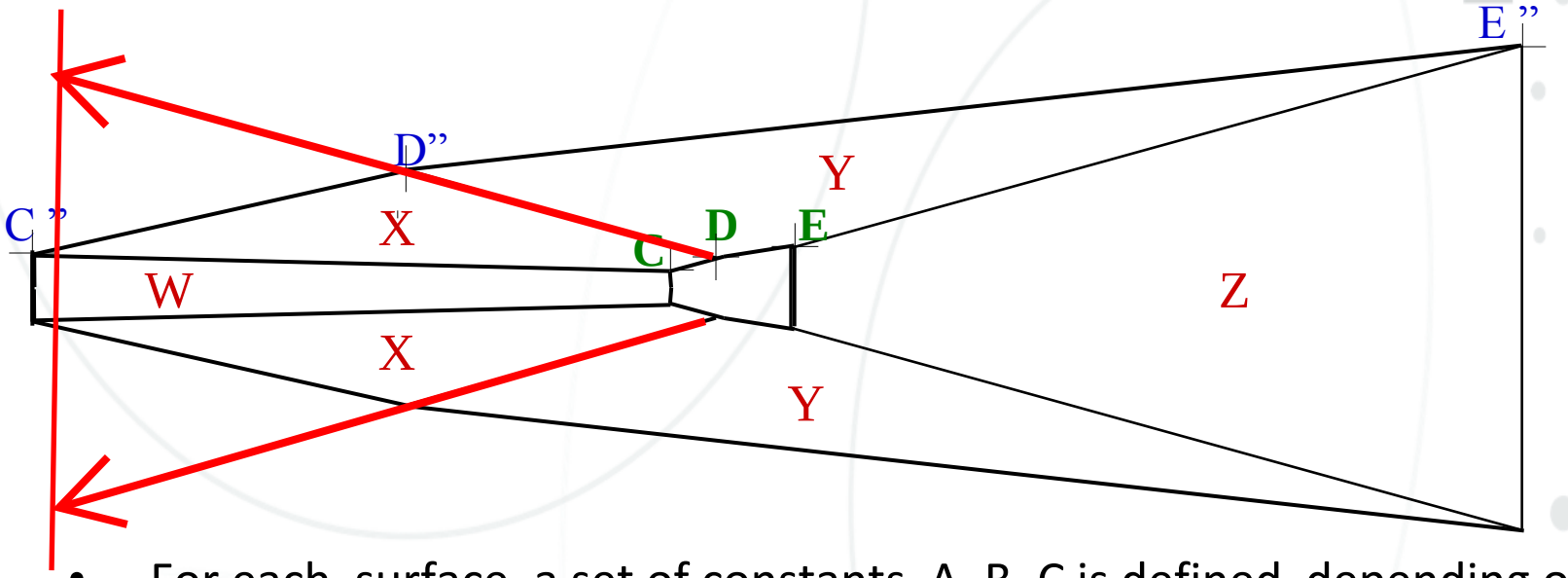
同ILS一致的

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Reminder :OAS Surfaces

- 6 sloping surfaces and a ground plane



- For each surface, a set of constants A, B, C is defined depending on
  - Distance Loc /THR: Distance between 2000 and 4500 m
  - GP (3° ,3.5°)
  - A/C dimensions,
  - Missed approach climb gradient,
  - RDH.

$$z = Ax+By+C$$

- For each point (x, y, z) the equation is :

⇨

Reminder : Summing up/ methodology

PAN-OPS

For each study if OAS is the selected method:

1- According to the parameters and necessary adjustements of coefficients input in the software, list the different equations and significant points of the OAS

2- Identify obstacles within the surfaces extended up to procedure altitude

3- Compare the height of the OAS surface with the height of the obstacle and list the penetrating obstacles

4- Identify if obstacles are approach or missed approach obstacles

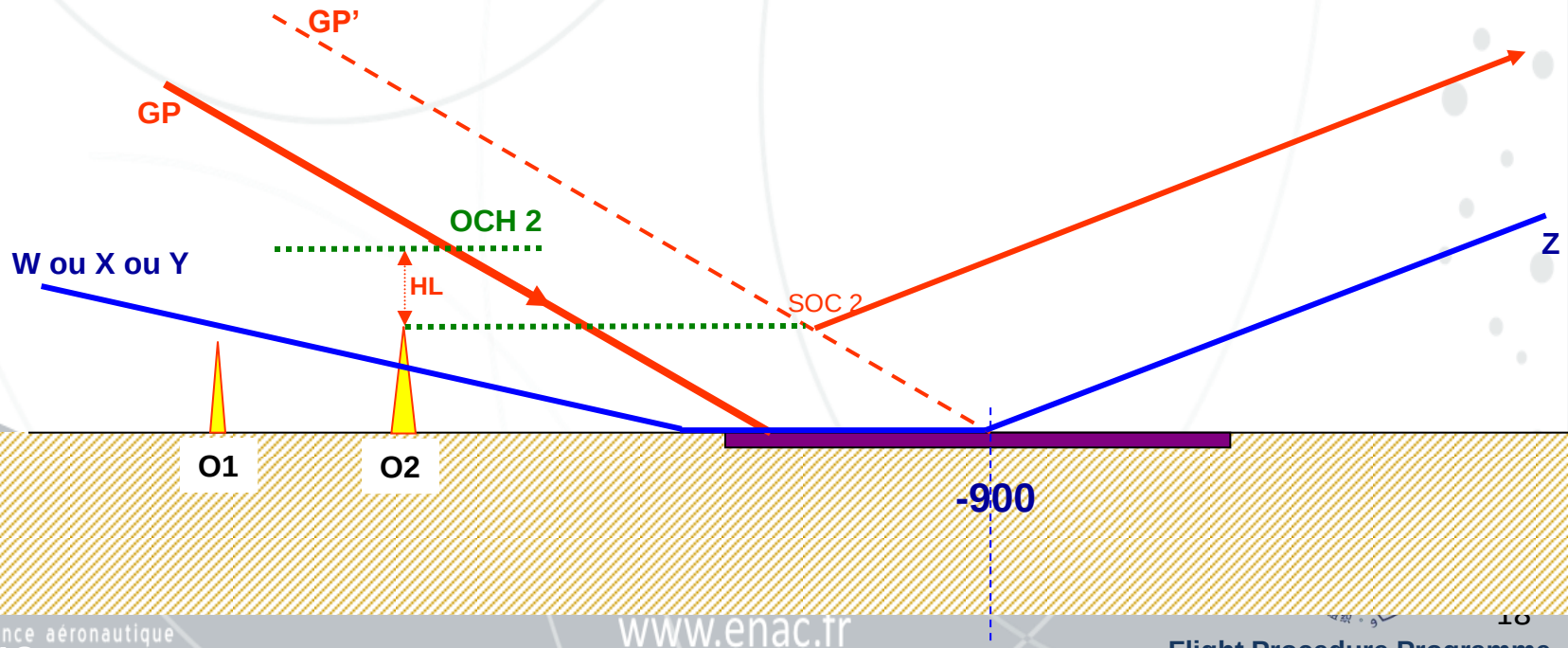
5- make adjustments to HL if necessary

6- Compute OCH

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Reminder : Computation of OCH for approach obstacles

- When obstacle penetrates W, X, Y or ground surface and do not penetrate GP' : **Hobst ≤ (x + 900) tan GP**
  - OCH = Hobst + HL
  - Xsoc = [(OCH-HL)-900] /tan GP



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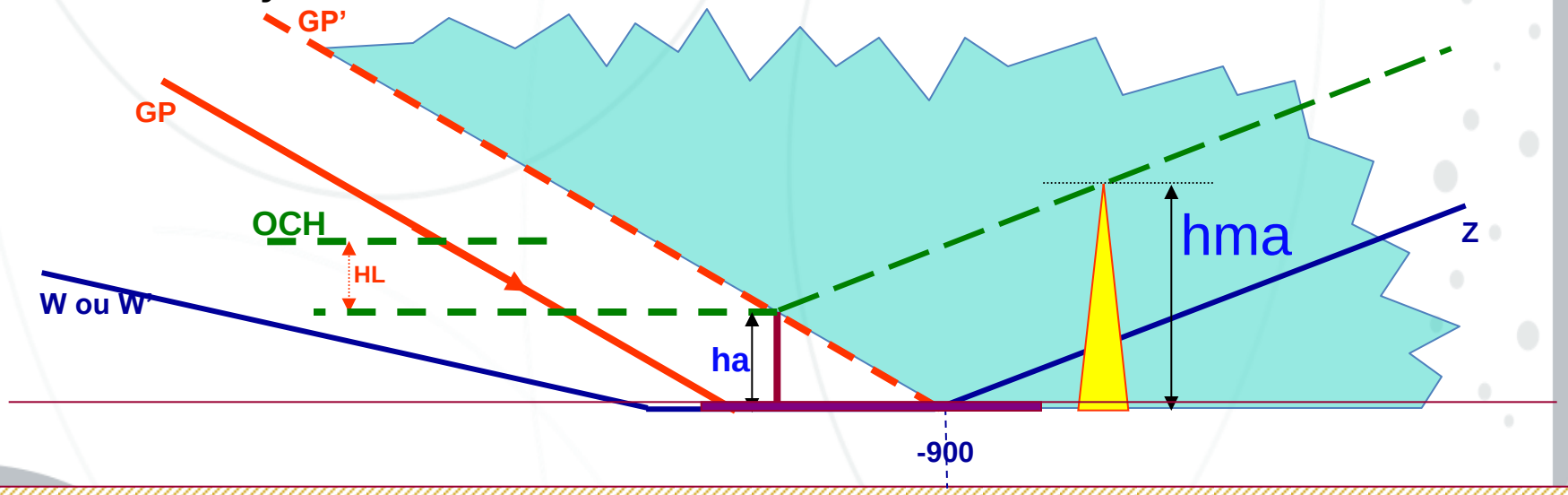
Reminder : Computation of OCH for Missed approach obstacles

2.5% : Cotan Z = 40

3° : Cotan α = 19

$$ha = \frac{hma \cot an Z + (xobs - XE)}{\cot an Z + \cot an \alpha}$$

The height of the equivalent obstacle called ha is the height of the SOC that enables to oberfly the obstacle hma



All obstacles penetrating GP' are considered as missed approach obstacles.

For missed approach obstacles , compute the equivalent obstacle Ha

OCH = ha+ HL

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INTERMEDIATE SEGMENT AREA

- Construction of the protection area:
- Intermediate area is the link between IF and FAF

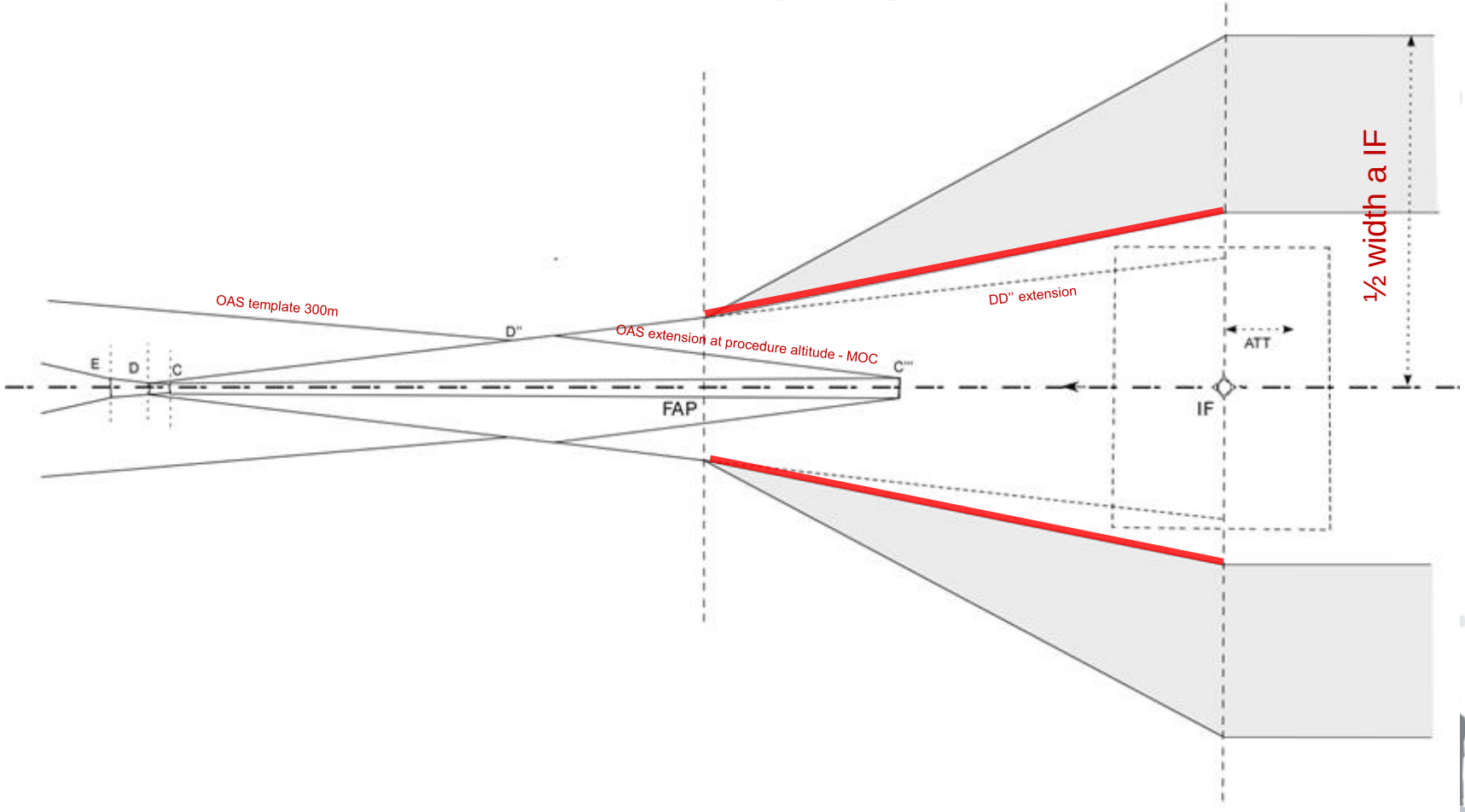
在IF点起始航段保护区宽度

  - At IF : ½ width of the end of initial RNAV segment
  - At FAP : width of X surface extension
  - Join the two areas

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INTERMEDIATE SEGMENT AREA

When X extension (DD'') is smaller than primary area at IF

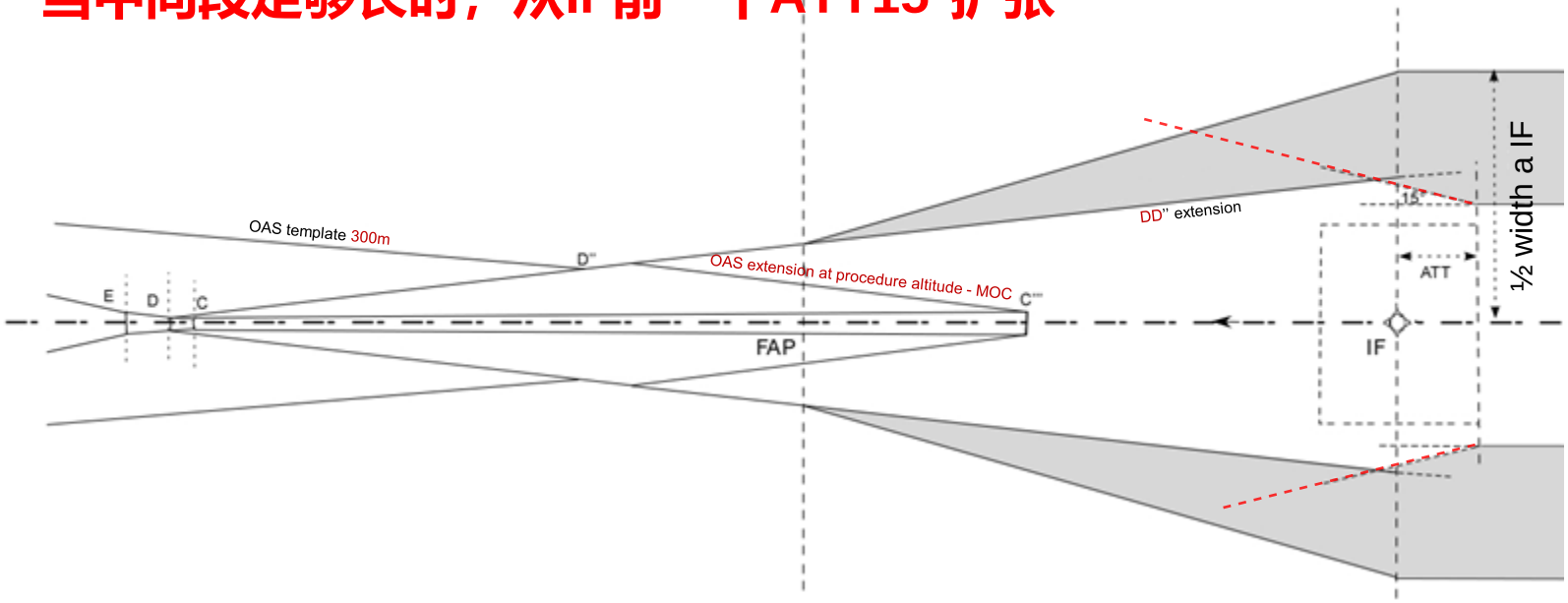


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INTERMEDIATE SEGMENT AREA

When X extension (DD’’) is wider than primary area at IF

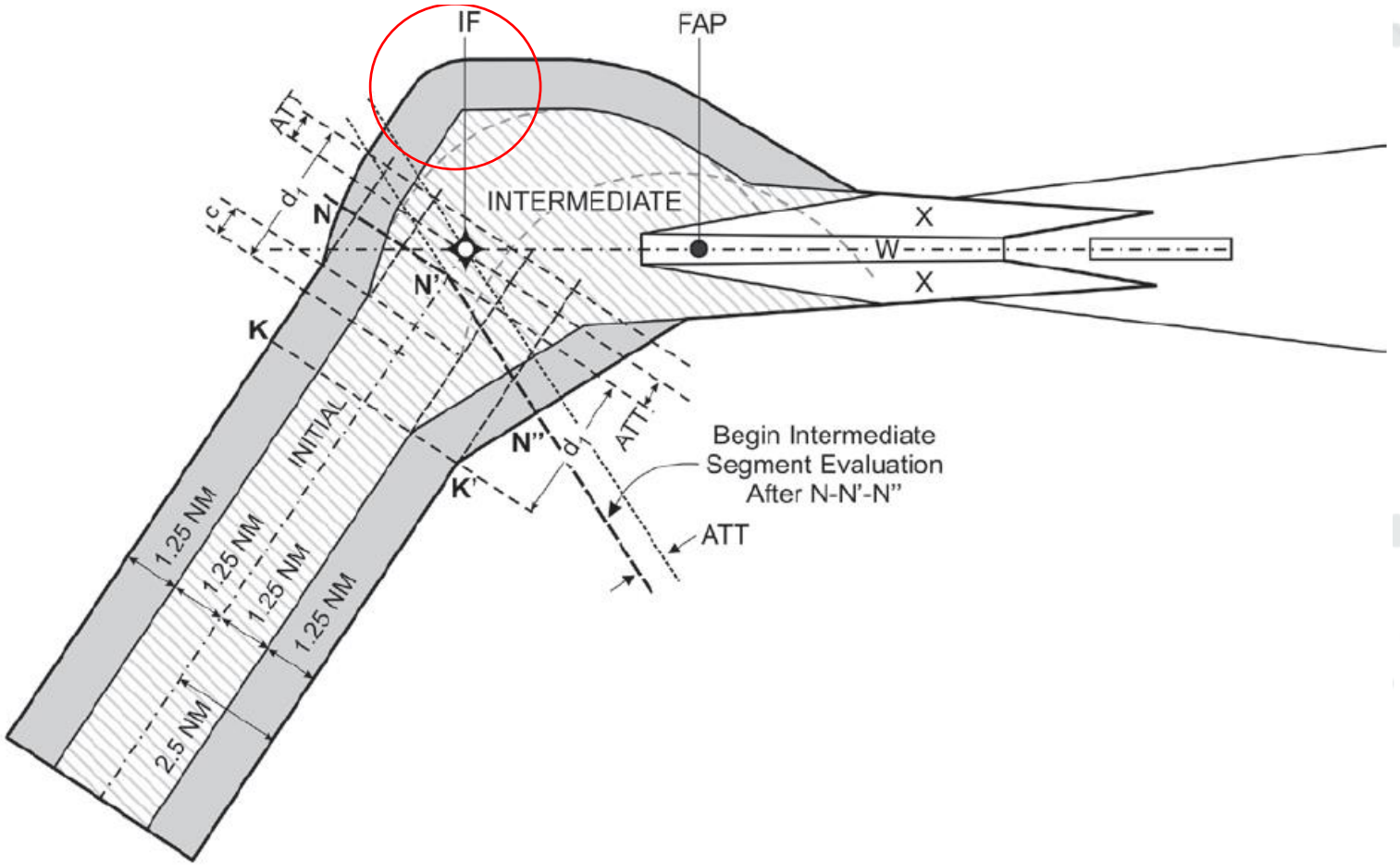
当中间段足够长时，从IF前一个ATT15°扩张



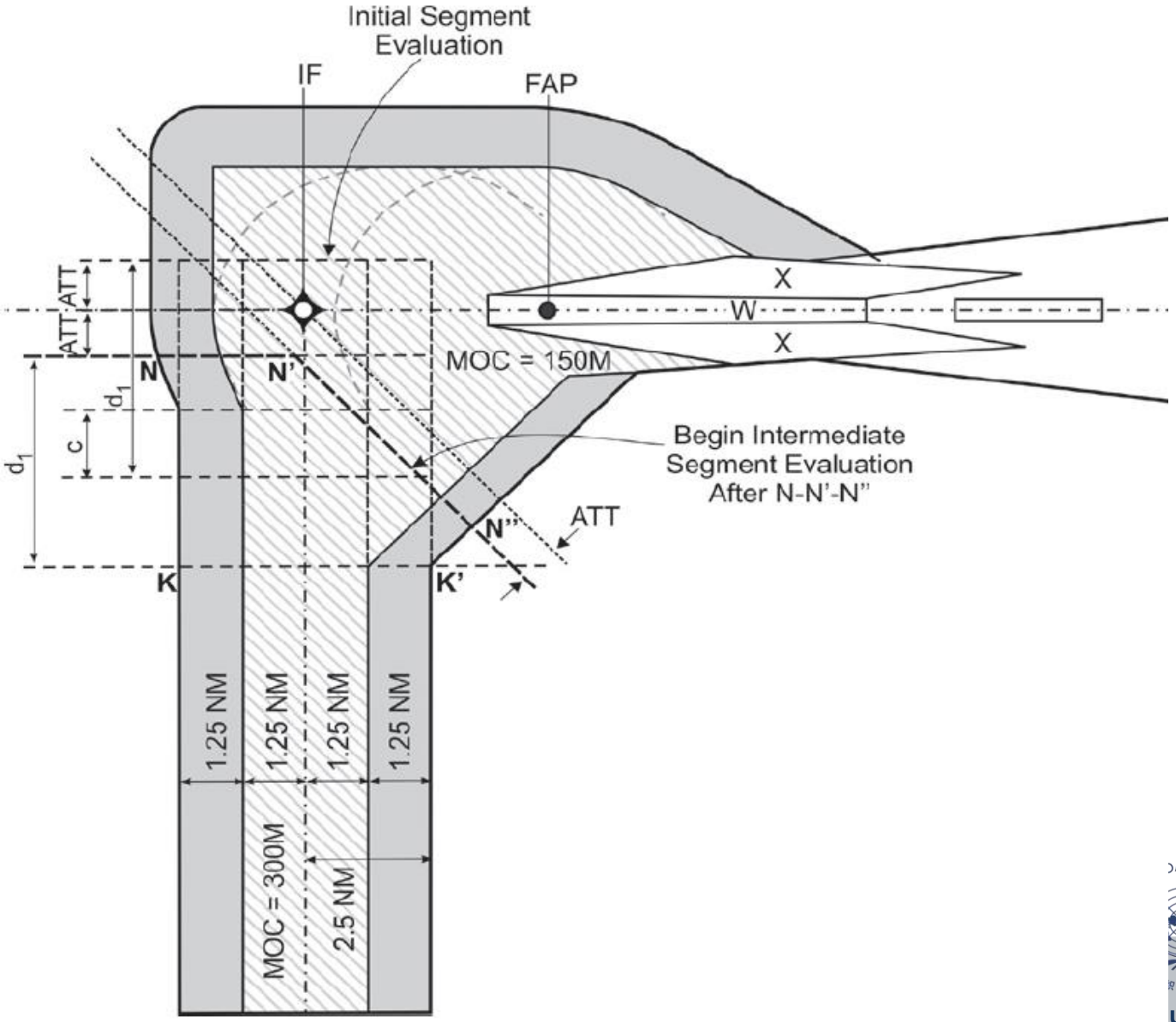
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Turn at IF : 60°

低速保护，和标称航迹平行



Turn at IF : 90°



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TRANSITION ILS TO RNAV

MISSED APPROACH SEGMENT

Know how to connect ILS precision segment to an RNAV missed approach

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Transition ILS to RNAV in missed approach

Ref II-1.5.1.4

•A missed approach can be based on RNAV or RNP navigation specifications.

•Only the systems capable of navigation accuracy of 1 NM or better in this phase of flight can be considered

•Turn at TP or turn at Altitude are possible

•RNAV guidance is considered as the latest from SOC

•Issue: the guidance is changing from accurate to less accurate as consequence the needle will remain centred even if the aircraft is diverging laterally from the track.

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Missed approach area : transition from ILS to RNAV

1. Draw protection area of RNAV/RNP missed approach from TP or TH and backward to SOC.

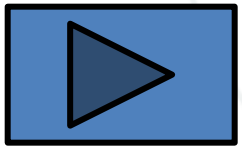
   1/2 AW = 2 Nm for RNP APCH, RNP1 or RNAV1

   extend扩张

2. Extend Z surface until reaching RNAV/RNP missed approach
3. Locate E' at the intersection between Z surface and RNAV/RNP missed approach primary area
4. At SOC, splay with OAS Y surface until reaching the RNAV/RNP missed approach secondary area

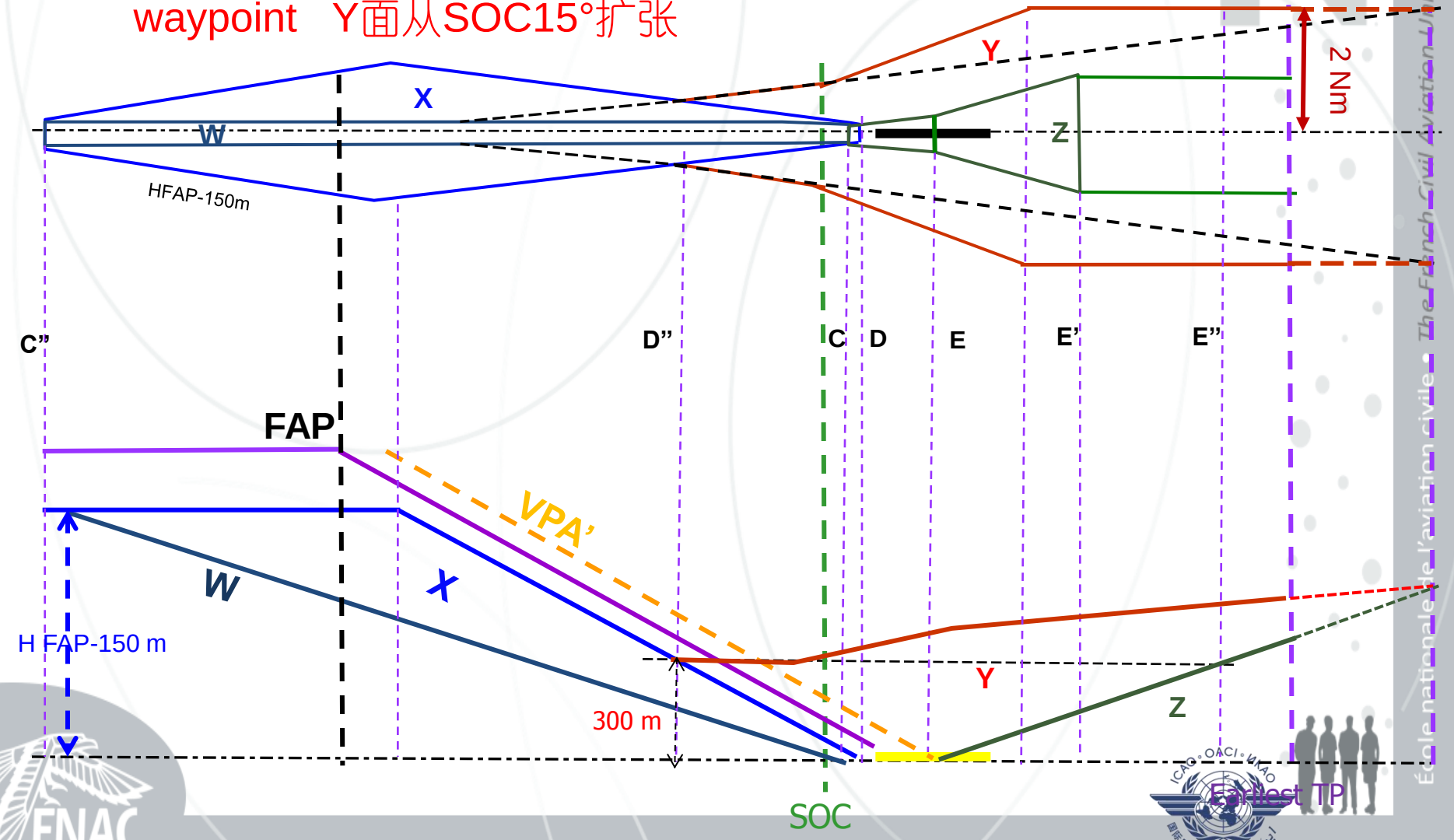
   NB : this is an iterative process as SOC location depends on OCH.

5. Until earliest TP or TH, the resulting surfaces RNAV/RNP (Primary or secondary) are considered as Y or Z surfaces extended laterally



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RNAV ILS Contours



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COMPUTE OCH

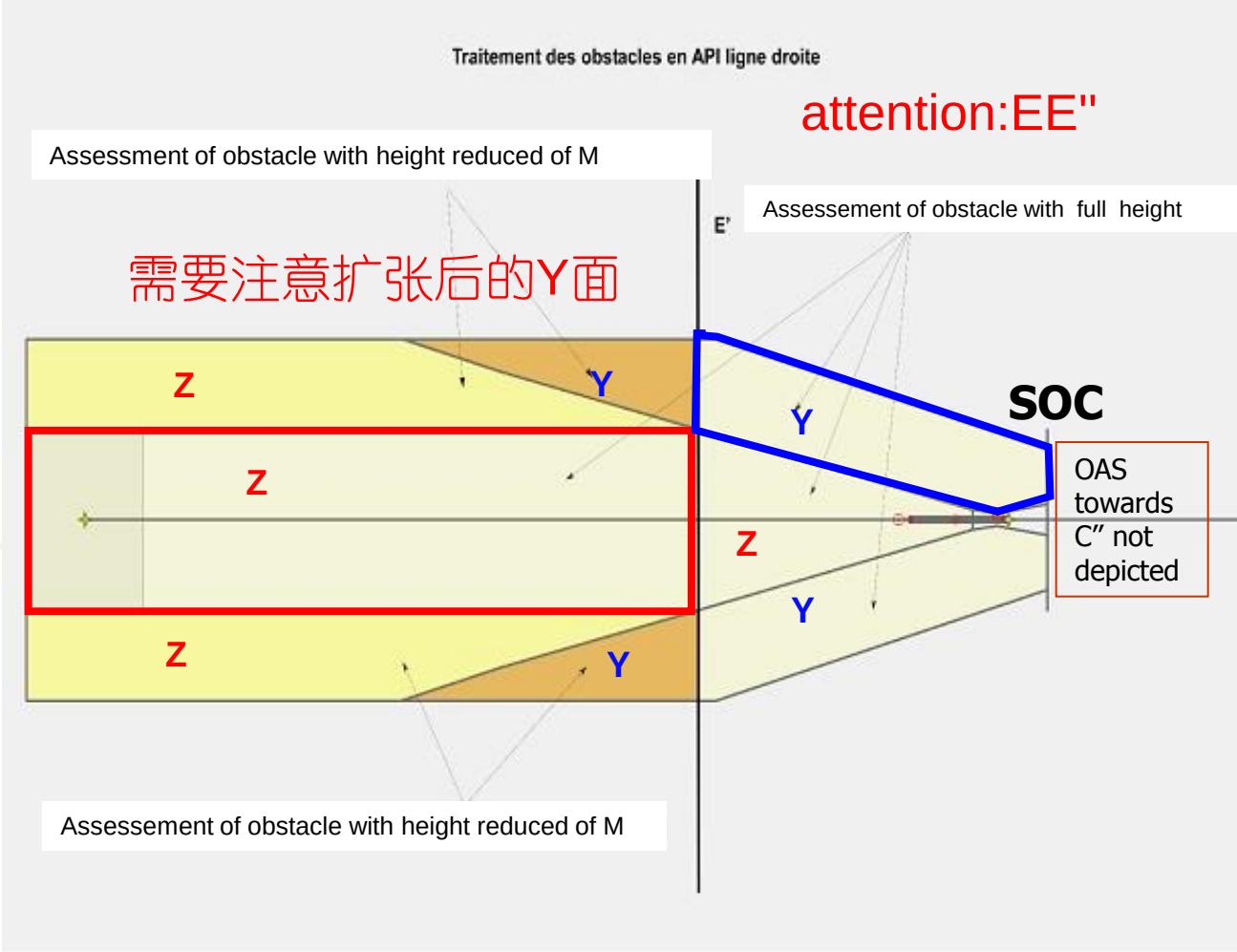
Straight in Missed approach

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Identification of obstacles

- Assessment of obstacle depends upon their position
- Between C'' and SOC
  - ILS OAS method
- Between SOC and E'
  - ILS OAS method
  - Special feature:lateral extension of Y surface
- Between E' and earliest TP or equivalent TH
  - Special assesment to take into account the guidance and the secondary area

EE"到TP，考虑M值



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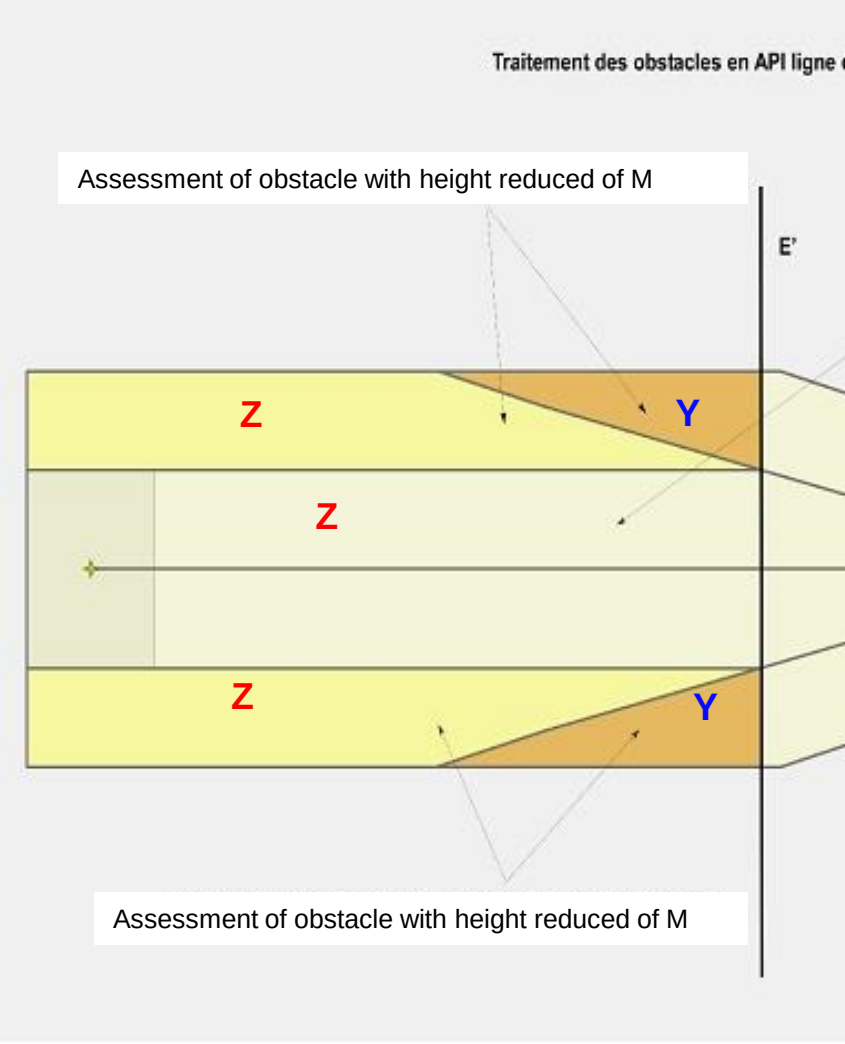
Obstacle assessment after E' and until earliest TP or equivalent TH

In RNAV/RNP primary area :

- Identify all obstacles penetrating Z surface
- Compute OCH. Full height is used for computation

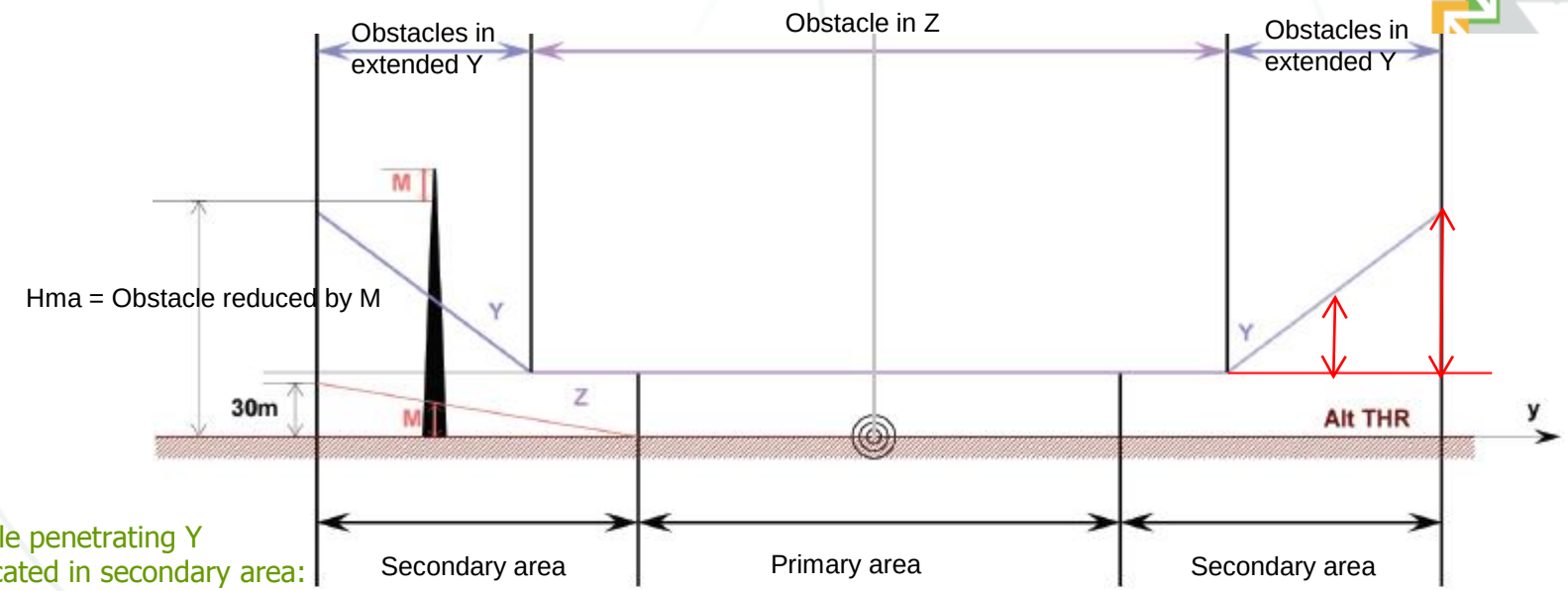
In RNAV/RNP secondary area

- Identify all obstacles penetrating Y and Z surfaces
- For those obstacles penetrating either Y or Z surfaces or their extensions within the secondary area, their heights are reduced by M
- M is zero at the edge of the primary area increasing linearly to 30 m (98 ft) at the edge of the total area. 主区边缘0m，副区边缘增加至30m（M)。
- Compute OCH. Reduced height is used for computation



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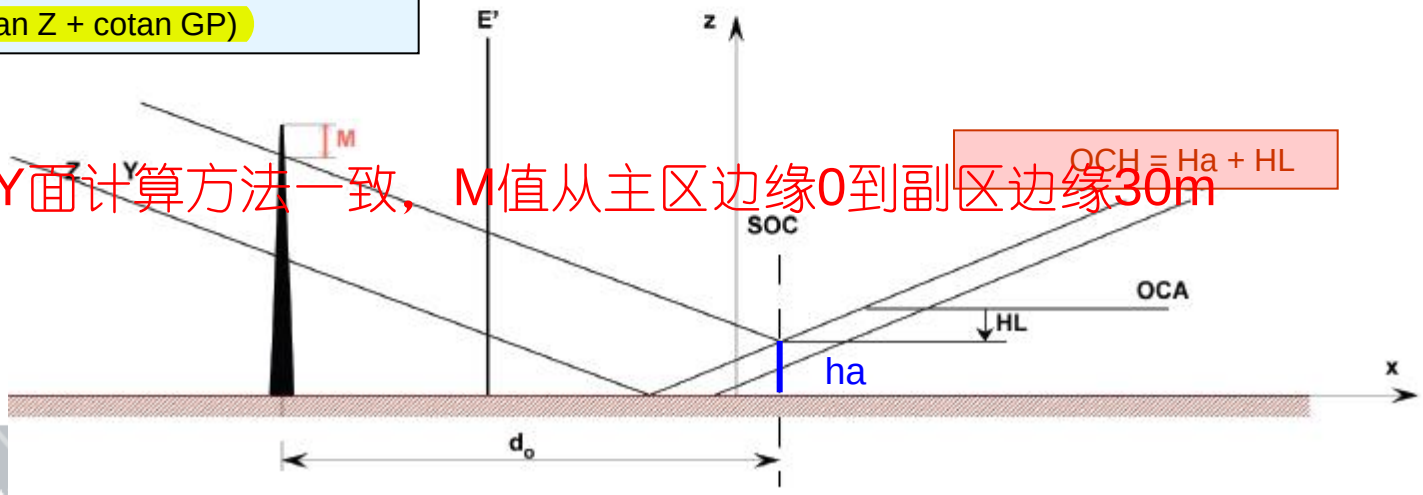
Sectional view (after point E')



Obstacle penetrating Y and located in secondary area:
Hma = (hobst – M)

$$ha = \frac{(hobst - M)\ \text{cotan}\ Z + (xobs + 900)}{(\text{cotan}\ Z + \text{cotan}\ GP)}$$

Profile view (after point E')

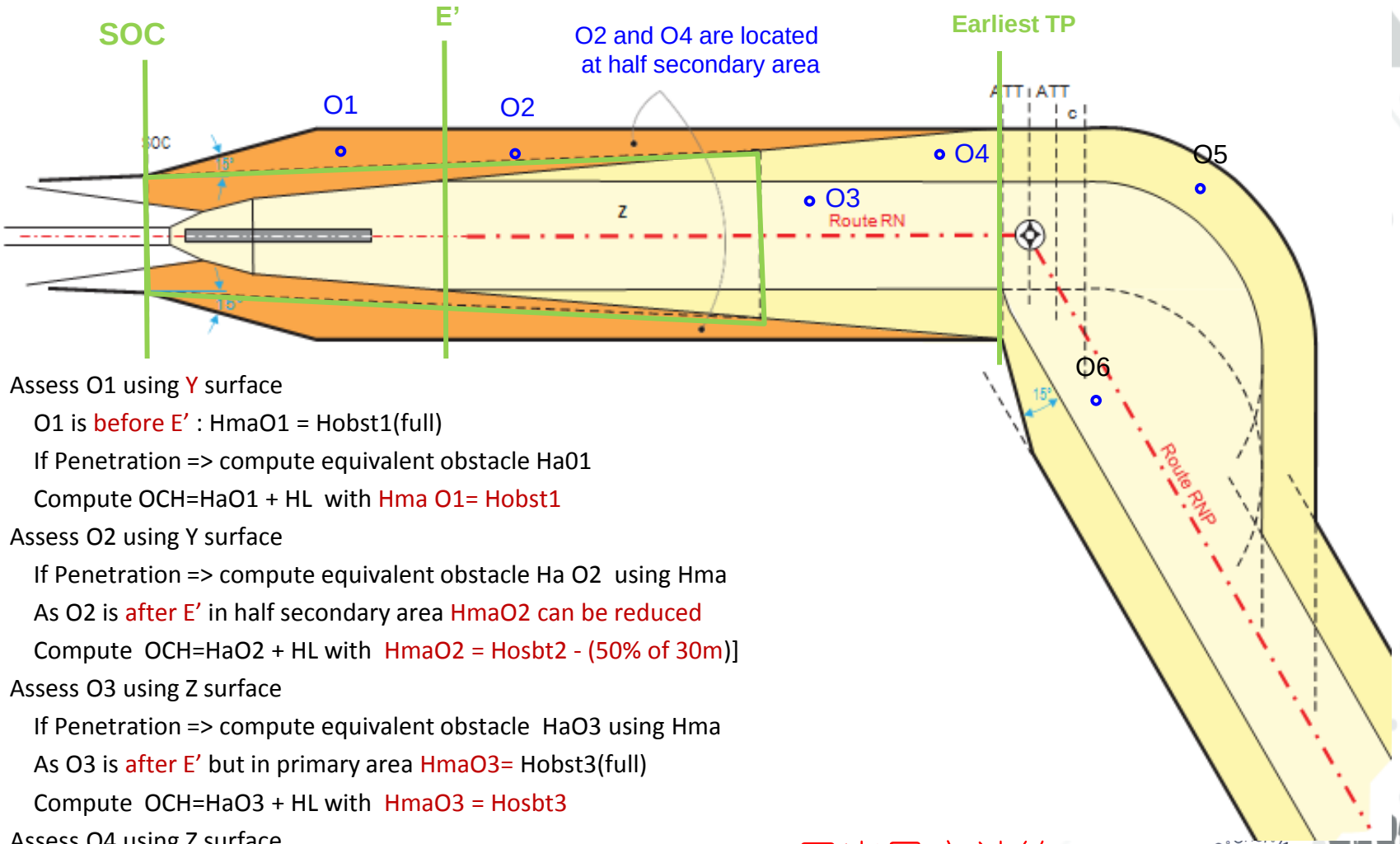


OCH = Ha + HL

Z面和Y面计算方法一致，M值从主区边缘0到副区边缘30m

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Obstacle assessment straight in missed approach Between SOC and earliestTP or equivalent TH



- Assess O1 using Y surface
  - O1 is before E' : HmaO1 = Hobst1(full)
  - If Penetration => compute equivalent obstacle Ha01
  - Compute OCH=HaO1 + HL with Hma O1= Hobst1
- Assess O2 using Y surface
  - If Penetration => compute equivalent obstacle Ha O2 using Hma
  - As O2 is after E' in half secondary area HmaO2 can be reduced
  - Compute OCH=HaO2 + HL with HmaO2 = Hosbt2 - (50% of 30m)]
- Assess O3 using Z surface
  - If Penetration => compute equivalent obstacle HaO3 using Hma
  - As O3 is after E' but in primary area HmaO3= Hobst3(full)
  - Compute OCH=HaO3 + HL with HmaO3 = Hosbt3
- Assess O4 using Z surface
  - If Penetration => computer HaO4
  - As O4 is after E' in half secondary area HmaO4 can be reduced
  - Compute OCH=HaO2 + HL with HmaO4 = Hosbt4 - (50% of 30m)]

用当量高计算OCH

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TRANSITION ILS TO RNAV

MISSED APPROACH SEGMENT WITH TP

Know how to connect ILS precision segment to an RNAV missed approach with a Turning point (TP)

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Special features for first Turning Point TP

- This waypoint can be Fly-by or Fly-over
- RF Not Permitted for first leg but can be used after a TF (defined on the LOC course)
- Max turn angle : 90°
- Earliest location of Fix
  - The minimum distance between SOC and first TP is:
  - *If TP is Fly over : ATT*
  - *If TP is Fly By : ATT + (rtan A/2 + 3s)*

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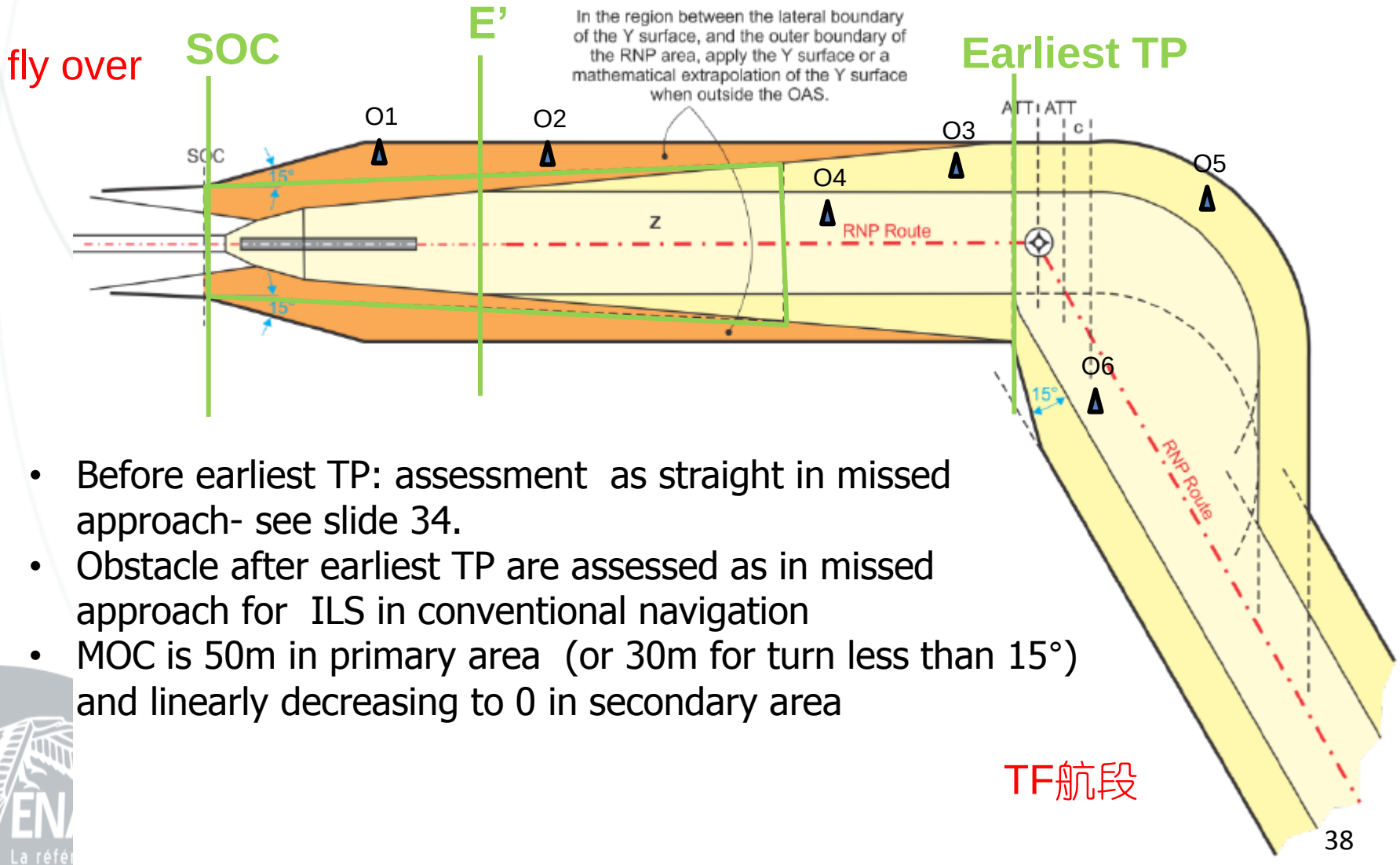
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COMPUTE OCH

ILS transition to RNAV/RNP Missed approach with a turning point TP

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Assessment of obstacles in the turn area



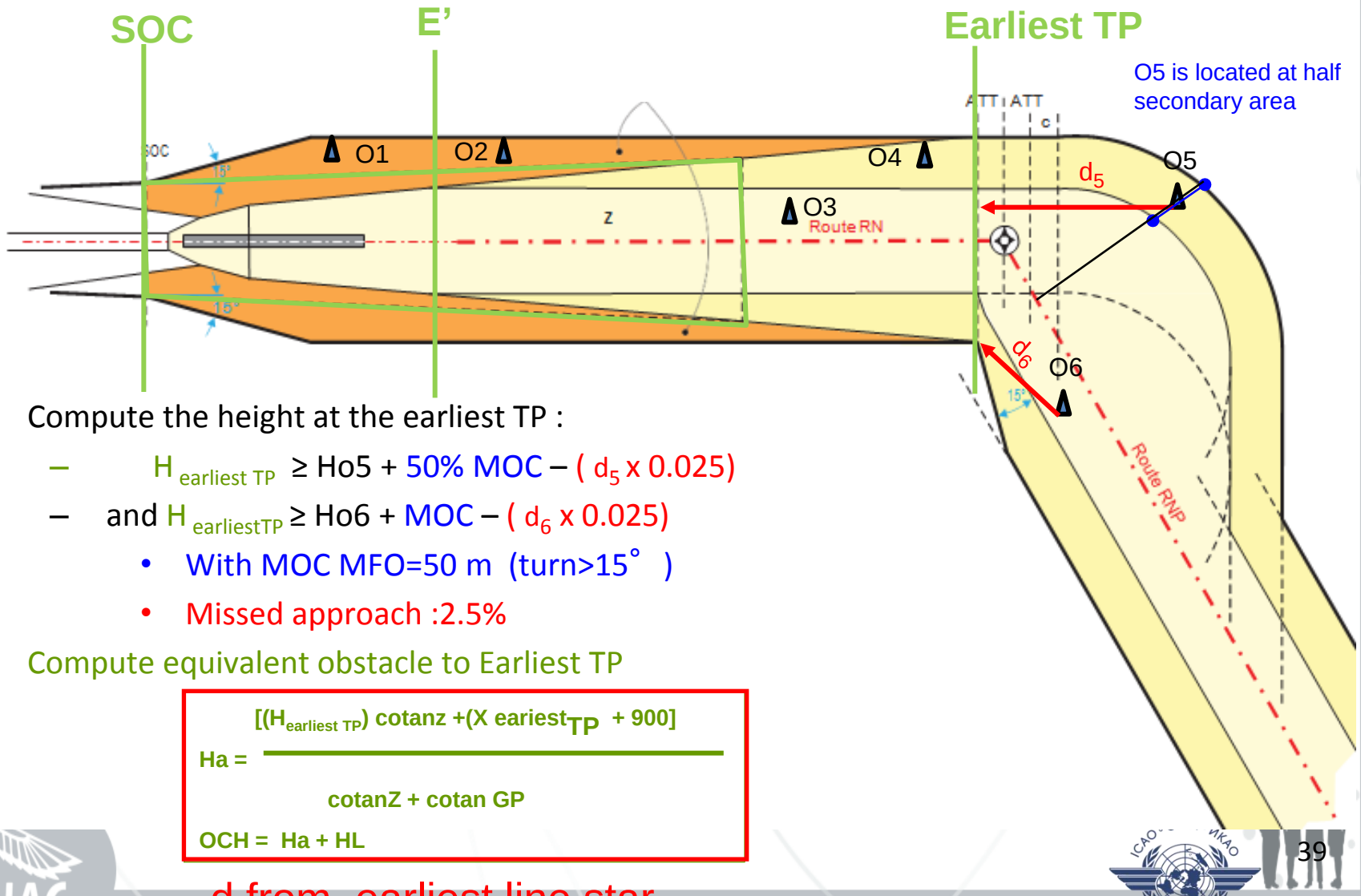
- Before earliest TP: assessment as straight in missed approach- see slide 34.
- Obstacle after earliest TP are assessed as in missed approach for ILS in conventional navigation
- MOC is 50m in primary area (or 30m for turn less than 15°) and linearly decreasing to 0 in secondary area

TF航段

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Assesment of obstacles in the turn area



- Compute the height at the earliest TP :
  - $H_{\text{earliest TP}} \geq Ho5 + 50\%\ MOC - (d_5 \times 0.025)$
  - and $H_{\text{earliestTP}} \geq Ho6 + MOC - (d_6 \times 0.025)$
    - With MOC MFO=50 m (turn>15°)
    - Missed approach :2.5%
- Compute equivalent obstacle to Earliest TP

$$Ha = \frac{[(H_{\text{earliest TP}})\ \text{cotanz} + (X\ \text{eariest}_{TP} + 900]}{\text{cotanZ} + \text{cotan GP}}$$

$$OCH = Ha + HL$$

d from earliest line star

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Fly By TP with Earliest TP before E'

- Until Earliest TP: Use straigtht in criteria :
  - Check if O1 is penetrating Y
  - If Yes compute equivalent obstacle HaO1 and OCH
- After earliest TP :Assess obstacle
  - within Y in the opposite side of the turn , If no penetration => obstacle are not considered for OCH computation
  - If penetrating Y and for all other obstacles in the turn area :

Compute Height at earliest TP :

$H_{earliest\ TP} \geq Ho2 + 50\%\ MOC - (d_2 \times 0.025)$

$H_{earliest\ TP} \geq Ho3 + MOC - (d_3 \times 0.025)$

Avec MOC=50 m (turn>15 °)

And missed climb gradient :2.5%

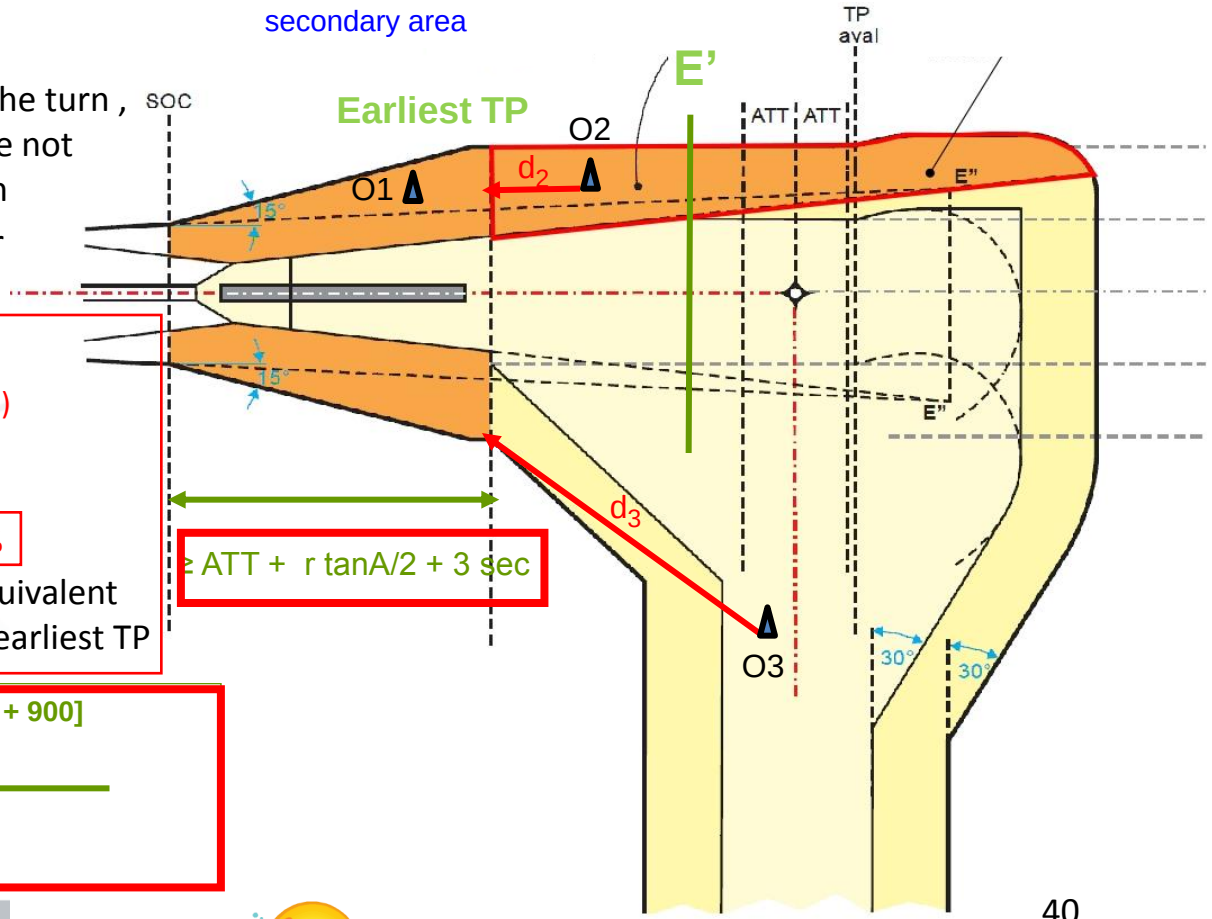
Use Ha formula to compute the equivalent obstacle of earliest TP with a height of Hearliest TP

$$Ha_{earliest\ TP} = \frac{[(H_{earliest\ TP})\ cotanz + (X_{earliest\ TP} + 900]}{cotanZ + cotan\ GP}$$

$$OCH = Ha_{earliest\ TP} + HL$$

O2 is located in the middle of secondary area

Obstacle below Y in the other side of turn might be not considered



The OCH is the higher OCH computed
Validate the position of SOC regarding the OCH



If SOC position and OCH do not match
⇨ Iterative process

Summing up
Assessment of Obstacles and OCH Computation

- Until the Point E' :
  - Use of OAS surface to compute OCH for penetrating obstacles as ILS
  - No secondary area and No reduced height of obstacle (M)

- After the point E' until earliest TP:
  - Primary area :
    - Use of Z surface and OCH computation for penetrating obstacles as in ILS.
    - No reduced height of obstacle (M)
  - Secondary area :
    - Identify obstacles penetrating Y, Z or their extensions
    - Reduce penetrating obstacles by M (linearly reduced from 30m at the edge of secondary area to 0 at the edge of primary area)
    - Compute Ha using Hma=Hobs –M
    - Compute OCH = Ha + HL

- After earliest TP (Turn area)
  - Obstacle not penetrating Y surface (from earliest TP) in the other side of the turn can be not considered for OCH computation.
  - Apply MOC 50m or 30 m(if turn angle≤15) in primary area and linearly decreasing in secondary area.
    - Compute Height at earliest TP :$H_{\text{earliest TP}} \geq Ho3 + MOC - (d3 \times 0.025)$
  - Compute Ha, the equivalent obstacle of earliest TP with hma= height of $H_{\text{earliest TP}}$
  - Compute OCH = Ha + HL

- OCH of the procedure is the highest OCH computed
- Validate the position of SOC regarding the OCH of the procedure
  - If SOC position and OCH do not match ⇨ Iterative process

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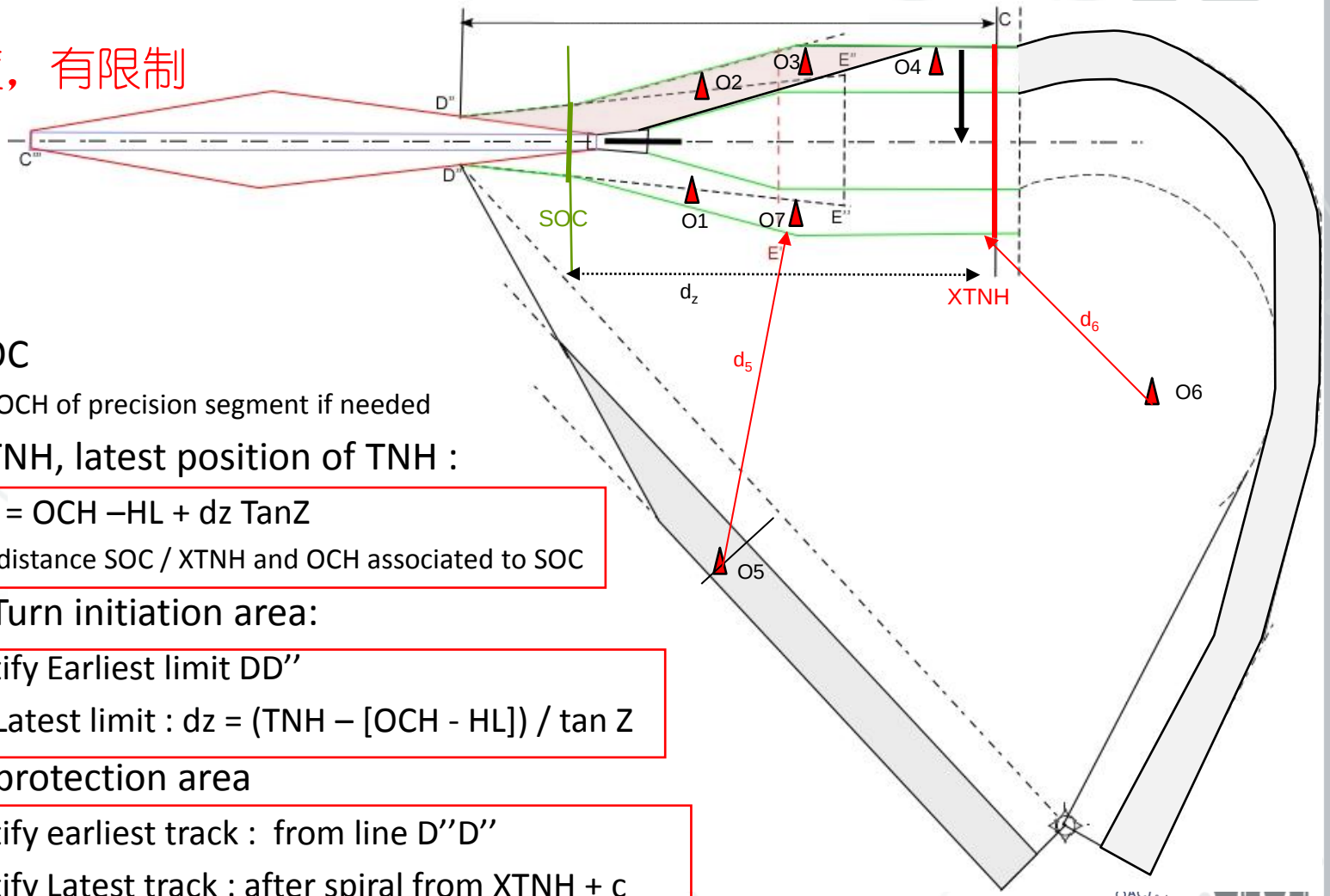
TRANSITION ILS RNAV

Connect ILS precision segment to an RNAV missed approach with a turn at an altitude and then DF to a waypoint

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Protection Area for a TH>300 m

2NM宽度，有限制



- Locate SOC
  - using OCH of precision segment if needed
- Locate XTNH, latest position of TNH :
  - TNH = OCH –HL + dz TanZ

  with dz = distance SOC / XTNH and OCH associated to SOC
- Idendity Turn initiation area:
  - Identify Earliest limit DD''
  - And Latest limit : dz = (TNH – [OCH - HL]) / tan Z
- Draw DF protection area
  - Identify earliest track : from line D''D''
  - Identify Latest track : after spiral from XTNH + c

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COMPUTE OCH

Missed approach with a turn at an altitude

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Turn at an altitude : assessment of obstacle

- Turn initiation area from D''D'' to dz (XTNH):
  - With dz = (TNH – [OCH - HL]) / tan Z and dz distance between SOC to XTNH
  - Assess obstacle as in straight area see slide 35:
  - Check TNH ≥ Ho + 50m for turn > 15° 转弯需要考虑ho+50m
  - Or check TNH ≥ Ho + 30 m for turn ≤ 15° 直线复飞
  - Obstacles within "Y" surface and not penetrating and located on the outer side of the turn need not be considered when calculating turn altitude/height
- Turn Area
  - Check TNH ≥ Ho + 50m - do tan z for turn > 15°
  - Or TNH ≥ Ho + 30 - do tan z for turn ≤ 15°
  - With do = shorter distance to turn initiation area

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Assessment of obstacles in the turn initiation area

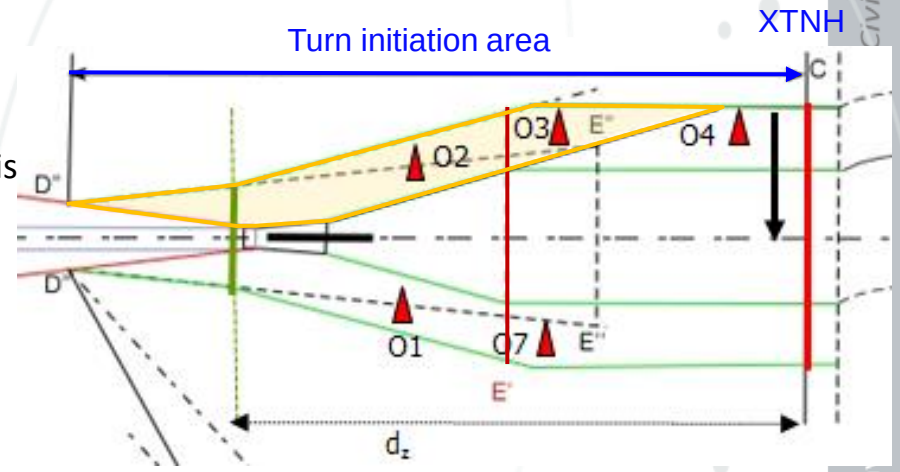
Objective : to compute OCH

- O1 and O2 are located before E' and within Y surface:
  - Are they penetrating? If yes :
    - Compute the equivalent obstacle Ha1 and Ha2 using the full height of O1 and O2
- O3, O4 and O7 are between E' and XTNH :
  - Assess O3 and O7 : O7 is penetrating Y surface and O3 is not
  - Assess O4 : it is penetrating Z surface
  - Compute equivalent obstacles for O4 and O7 :

  Ha7 et Ha4 using Hma =[Hobstacle –(50% x M)]

  With M=30m and considering all obstacles are located at half RNAV secondary area
- Compute OCH = max(Ha) + HL
- Depending on the result :
  - Check SOC position and 15° splay
  - Check XTNH from new SOC position if needed

穿透



EE'之前，考虑全高度的当量高，之后主副区，考虑M值

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Obstacle assessment

Turn area including turn initiation area

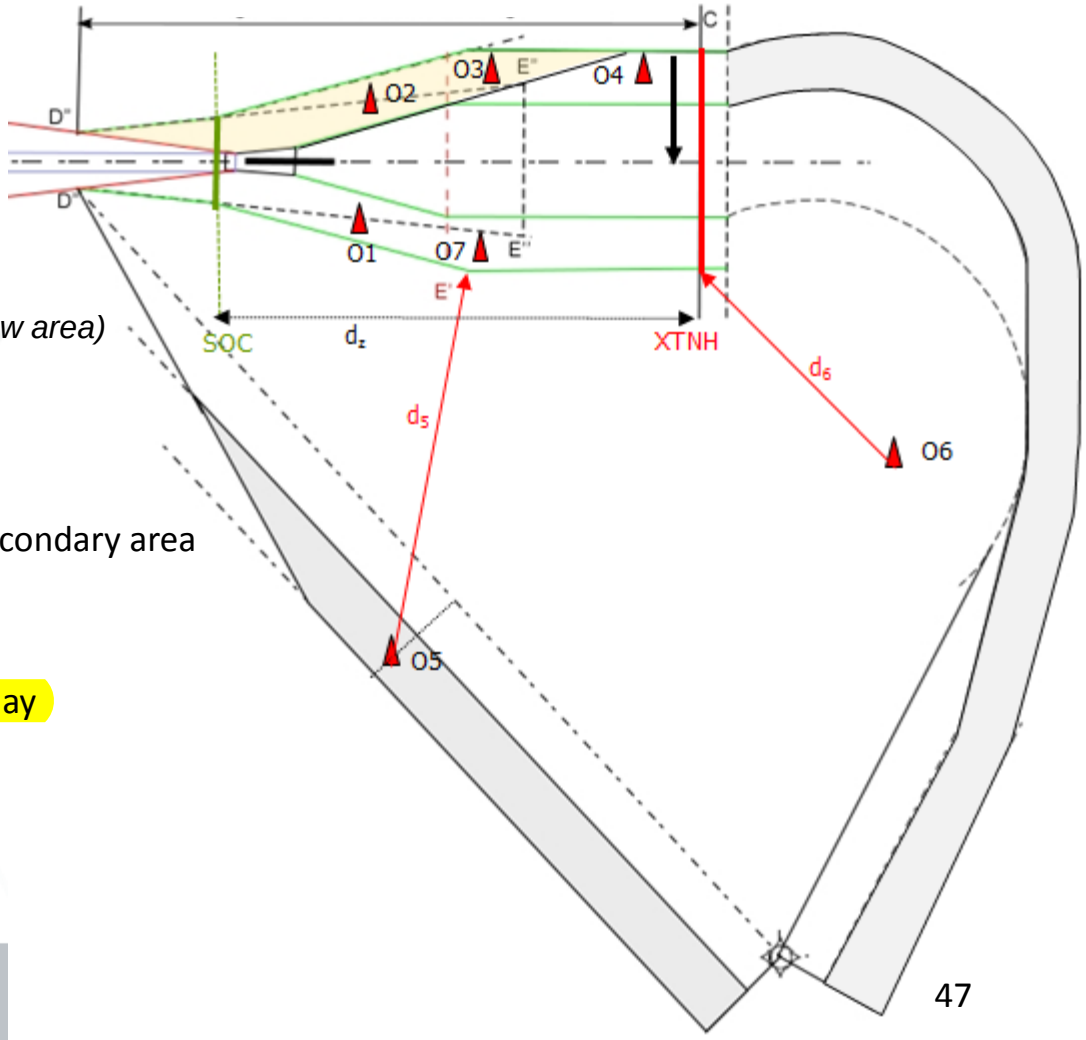
Objective : to validate the turning height (TNH)

1. In Turn initiation Area, check :
   - TNH ≥ Ho1 + 50 m
   - TNH ≥ Ho2 + 50 m
   - TNH ≥ Ho7 + 50 m
   - TNH ≥ Ho4 + 50 m

TNH is not checked for O3 as it is not penetrating Y (yellow area)

2. In turn area, check:
   - TNH ≥ Ho6 + 50 m – (d6 x 0.025)
   - TNH ≥ Ho5 + (50% x 50 m)– (d5 x 0.025)
     - O5 is located in the middle of RNAV secondary area

3. Depending on the result, increase TNH:
   - Without moving XTNH => increase OCH
   - move SOC location => move origin of 15° splay
   - move XTNH forward without increasing OCH
   - Increase both OCH and XTNH



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MISSED APPROACH WITH A TURN AT AN ALTITUDE

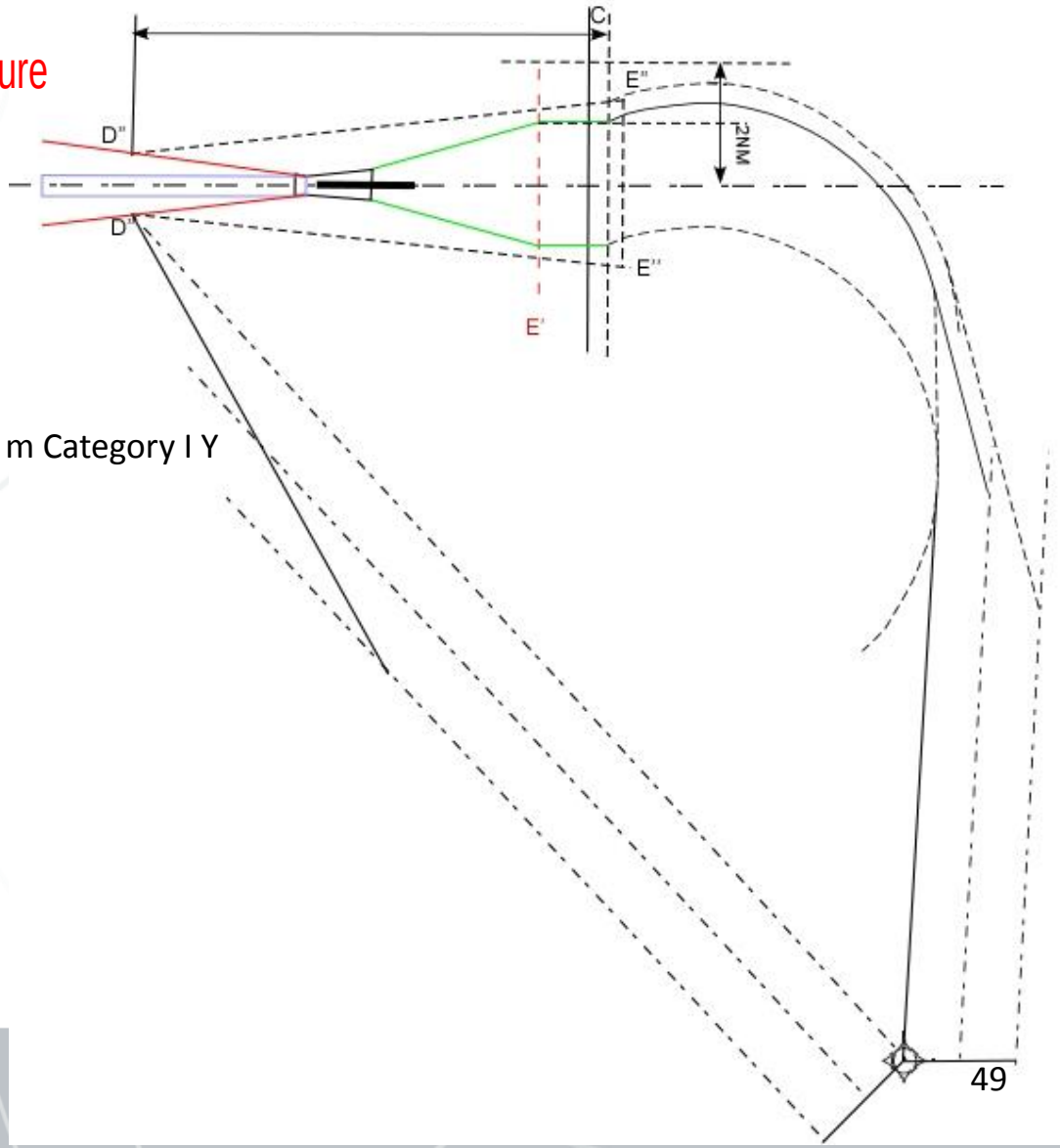
Turn at TH ≤ 300m

Turn as soon as practicable

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Protection Area for a TH< 300 m

2NM宽度，TH<300m,we can use this picture



The turn initiation area is bounded by the 300 m Category I Y surface contour, and terminates at the TP

- NO splay from SOC
- Extended Y surface does'nt exist
- Obstacle assesment as in slide 46

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Turn as soon as practicable

- Turn initiation area
  - Compute OCH Ps
  - Earliest limit DD''
  - Latest limit SOC: dz=0
  - Check TNH ≥ Ho + 50m for turn > 15°
  - Or check TNH ≥ Ho + 30 m for turn ≤ 15
  - Check TNH= OCHps –HL
- Turn area
  - Check TNH ≥ Ho + 50m - do tan z for turn > 15°
  - Or TNH ≥ Ho + 30 for turn ≤ 15°
- The chart shall be annotated "turn as soon as practicable to…''

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PUBLICATION

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- ILS RWY XX
- RNAV/RNP application in the PBN Box.

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